R Programming Practical Examination

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Part A: Basic Operations

A1

```
v <- seq(5,50, by = 5)
v
```

```
##  [1]  5 10 15 20 25 30 35 40 45 50
```

```
length(v)
```

```
## [1] 10
```

```
sum(v)
```

```
## [1] 275
```

```
mean(v)
```

```
## [1] 27.5
```

A2

```
w <- rep(c(1,2,3), times = 4)
z <- rep(c(1,2,3), each = 4)

w
```

```
##  [1] 1 2 3 1 2 3 1 2 3 1 2 3
```

```
z
```

```
##  [1] 1 1 1 1 2 2 2 2 3 3 3 3
```

A3

```
rem <- 17%%5
int_div <- 17 %/% 5
pow <- 2 ^ 10
sq <- sqrt(144)
pi
```

```
## [1] 3.141593
```

```
rem
```

```
## [1] 2
```

```
int_div
```

```
## [1] 3
```

```
pow
```

```
## [1] 1024
```

```
sq
```

```
## [1] 12
```

```
pi
```

```
## [1] 3.141593
```

A4

```
a <- c(2,4,6,8)
b <- c(1,3,5,7)

plus <- a + b
prod <- a * b
logical_comparison <- a > b
dot_prod <- sum(a * b)

plus
```

```
## [1]  3  7 11 15
```

```
prod
```

```
## [1]  2 12 30 56
```

```
logical_comparison
```

```
## [1] TRUE TRUE TRUE TRUE
```

```
dot_prod
```

```
## [1] 100
```

A5

```
## paste()

a <- 42
b <- 42L
c <- "42"
d <- 41i
e <- TRUE

typeof(a)
```

```
## [1] "double"
```

```
typeof(b)
```

```
## [1] "integer"
```

```
typeof(c)
```

```
## [1] "character"
```

```
typeof(d)
```

```
## [1] "complex"
```

```r
typeof(e)
```

```
## [1] "logical"
```

Part B: Data Frames and Conditional Statements

B1

```r
students <- data.frame(
  id = 1:8,
  name = c("Alice", "Bruno", "Chen", "Dana", "Elif", "Farid", "Grace", "Hiro"),
  sex = c("F", "M","M","F","F","M","F","M"),
  section = c("A", "B", "A", "C", "B", "C", "A", "B"),
  midterm = c(78, 85, 92, 67, 74, 88, 95, 59),
  final = c(82, 79, 96, 71, 80, 91, 93, 65),
  attendance = c(95, 88, 100, 72, 90, 85, 98, 60),
  stringsAsFactors = FALSE
)
students
```

```
##   id  name sex section midterm final attendance
## 1  1 Alice   F       A      78    82         95
## 2  2 Bruno   M       B      85    79         88
## 3  3  Chen   M       A      92    96        100
## 4  4  Dana   F       C      67    71         72
## 5  5  Elif   F       B      74    80         90
## 6  6 Farid   M       C      88    91         85
## 7  7 Grace   F       A      95    93         98
## 8  8  Hiro   M       B      59    65         60
```

B2

```r
str(students)
```

```
## 'data.frame':    8 obs. of  7 variables:
##  $ id        : int  1 2 3 4 5 6 7 8
##  $ name      : chr  "Alice" "Bruno" "Chen" "Dana" ...
##  $ sex       : chr  "F" "M" "M" "F" ...
##  $ section   : chr  "A" "B" "A" "C" ...
##  $ midterm   : num  78 85 92 67 74 88 95 59
##  $ final     : num  82 79 96 71 80 91 93 65
##  $ attendance: num  95 88 100 72 90 85 98 60
```

```r
dim(students)
```

```
## [1] 8 7
```

```r
ncol(students)
```

```
## [1] 7
```

```r
summary(students)
```

```
##        id             name               sex             section     
##  Min.   :1.00   Length   :8   Length   :8   Length   :8   Min.   :59.00  
##  1st Qu.:2.75   N.unique :8   N.unique :2   N.unique :3   1st Qu.:72.25  
##  Median :4.50   N.blank  :0   N.blank  :0   N.blank  :0   Median :81.50  
##  Mean   :4.50   Min.nchar:4   Min.nchar:1   Min.nchar:1   Mean   :79.75  
```

```
##  3rd Qu.:6.25   Max.nchar:5   Max.nchar:1   Max.nchar:1   3rd Qu.:89.00
##  Max.   :8.00                                             Max.   :95.00
##      final         attendance
##  Min.   :65.00   Min.   : 60.00
##  1st Qu.:77.00   1st Qu.: 81.75
##  Median :81.00   Median : 89.00
##  Mean   :82.12   Mean   : 86.00
##  3rd Qu.:91.50   3rd Qu.: 95.75
##  Max.   :96.00   Max.   :100.00
```

B3

```
students$average <- round((students$midterm + students$final) / 2, 1)
students$average
```

```
## [1] 80.0 82.0 94.0 69.0 77.0 89.5 94.0 62.0
```

B4

```
students$passed <- (students$average >= 70) & (students$attendance >= 75)
students$passed
```

```
## [1]  TRUE  TRUE  TRUE FALSE  TRUE  TRUE  TRUE FALSE
```

B5

```
students$standing <- ifelse(students$average >= 90, "Honors",
ifelse(students$average >= 75, "Good", "Needs Support"))
students$standing
```

```
## [1] "Good"          "Good"          "Honors"        "Needs Support"
## [5] "Good"          "Good"          "Honors"        "Needs Support"
```

B6

```
classify <- function(score) {
  if (score >= 85){
   return("Disfunction")
  } else if (score >= 70){
     return("Merit")
  } else if (score >= 50){
    return ("Pass")
  } else {
      return("Fail")
    }
}
sapply(c(90, 73, 55, 40), classify)
```

```
## [1] "Disfunction" "Merit"       "Pass"        "Fail"
```

Part C: rbind() and cbind()

C1

```
new_students <- data.frame(
  id = c(9, 10),
  name = c("Ines", "Jamal"),
  sex = c("F", "M"),
  section = c("C", "A"),
```

```
  midterm = c(81, 70),
  final = c(84, 68),
  attendance = c(92, 80),
  stringsAsFactors = FALSE
)

new_students$average <- round((new_students$midterm + new_students$final) / 2, 1)
new_students$passed <- (new_students$average >= 70) & (new_students$attendance >= 75)
new_students$standing <- ifelse(new_students$average >= 90, "Honors", ifelse(new_students$average >= 75

new_students
```

```
##   id  name sex section midterm final attendance average passed      standing
## 1  9  Ines   F       C      81    84         92    82.5   TRUE          Good
## 2 10 Jamal   M       A      70    68         80    69.0  FALSE Needs Support
```

```
new_students$average
```

```
## [1] 82.5 69.0
```

```
new_students$passed
```

```
## [1]  TRUE FALSE
```

```
new_students$standing
```

```
## [1] "Good"          "Needs Support"
```

```
students_all <- rbind(students, new_students)
students_all
```

```
##    id  name sex section midterm final attendance average passed      standing
## 1   1 Alice   F       A      78    82         95    80.0   TRUE          Good
## 2   2 Bruno   M       B      85    79         88    82.0   TRUE          Good
## 3   3  Chen   M       A      92    96        100    94.0   TRUE        Honors
## 4   4  Dana   F       C      67    71         72    69.0  FALSE Needs Support
## 5   5  Elif   F       B      74    80         90    77.0   TRUE          Good
## 6   6 Farid   M       C      88    91         85    89.5   TRUE          Good
## 7   7 Grace   F       A      95    93         98    94.0   TRUE        Honors
## 8   8  Hiro   M       B      59    65         60    62.0  FALSE Needs Support
## 9   9  Ines   F       C      81    84         92    82.5   TRUE          Good
## 10 10 Jamal   M       A      70    68         80    69.0  FALSE Needs Support
```

C2

```
quiz <- c(8,7,10,6,7,9,10,5,8,6)
students_all <- cbind(students_all, quiz)
class(students_all)
```

```
## [1] "data.frame"
```